INCH POUND

MIL-C-3098/138D
23 December 1991
SUPERSEDING
MIL-C-3098/138C
2 November 1983

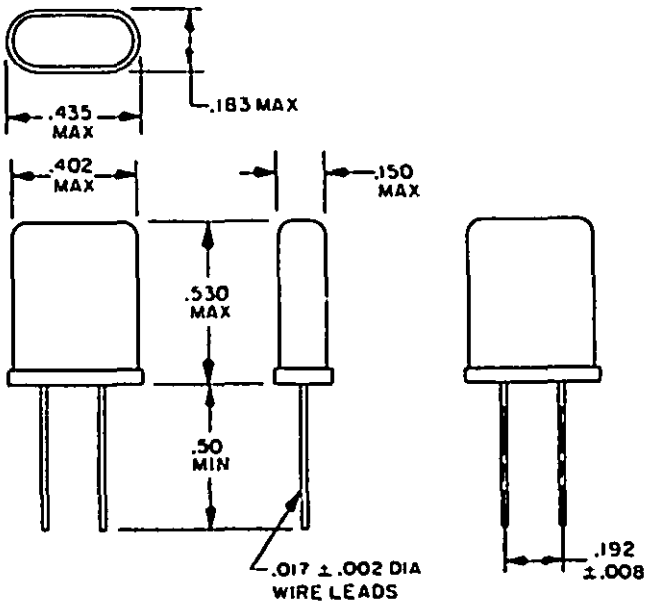
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR154/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent Characteristics: 17 to 61 MHz; third overtone; controlled; series.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR154/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/09) or HC-43/U (MIL-H-10056/30).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 to 61 MHz, inclusive.

Mode of oscillation: Third mechanical overtone.

Reference temperature: +85°C ±1°C.

Operating temperature range (controlled): +80°C to +90°C, inclusive.
Frequency tolerance: ± 20 ppm.
Equivalent resistance: 40 ohms, maximum.

Frequency stability: ± 5 ppm.

Operable temperature range: -55°C to 80°C, inclusive.

Unwanted modes: Method II, minimum unwanted-to-main mode resistance ratio of 2:1 or 40 ohms, whichever is greater.

Drive level:
17 to 25 MHz: 2.0 mW ±0.4 mW.
25+ to 61 MHz: 1.0 mW ±0.2 mW.

Capacitance, shunt: 4.0 pf, maximum.

Shock:
Frequency change permitted: ±5 ppm.
Equivlent resistance change permitted: ±10 percent.

Vibration: Method 201 Of MIL-STD-202
Frequency change permitted: ±5 ppm.
Equivlent resistance change permitted: ±10 percent.

Temperature cycling:
Frequency change permitted: ±5 ppm.
Equivalent resistance change permitted: ±10 percent.

Aging:
Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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